Active Directory Identity & Access Troubleshooting Lab

A home-lab project in a **simulated 200-user healthcare Active Directory environment**. I built the environment and resolved a realistic identity/access incident (Ticket NMG-0047) end-to-end — from diagnosis through resolution, verification, and prevention.

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1. Problem Statement

A newly onboarded Operations Coordinator (Jane Cooper) could log in successfully but received **Access Denied** on all Operations shared resources. Her desktop environment — group policies, mapped drives, and restrictions — also behaved differently from the rest of her team, while no other Operations team member was affected. The issue persisted after multiple restarts.

Left unresolved, this blocked a new hire from working during her first week and pointed to a possible gap in the account-provisioning process.

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2. Solutions Overview

I resolved the issue by comparing the affected account against a known-good teammate (Carlos Rivera) to isolate exactly what was different — investigating fully **before** making any change.

The comparison surfaced **two separate root causes** from a single sloppy account setup:

- **Wrong Organizational Unit (OU):** Jane was placed in the **HR OU** instead of the Operations OU, so the wrong Group Policy applied to her account (explaining the different desktop, drives, and policies).
- **Missing security group:** Jane was not a member of the `Operations-Users` group, which is what grants access to Operations resources (explaining the Access Denied). She had also been added to `HR-Users` in error.

Fixes applied (one per root cause):

1. Moved Jane from the HR OU to the **Operations OU** so the correct Group Policy applies.
2. Added Jane to the `Operations-Users` security group to restore resource access.
3. Removed Jane from `HR-Users` to eliminate unintended HR access.

Verified both symptoms cleared — correct OU, correct group membership, correct policies applied after `gpupdate /force` and re-login, and confirmed access to Operations resources.

> **Key lesson:** The obvious symptom (Access Denied) hid a second root cause (OU/GPO). Fixing only the visible one would have left the ticket half-resolved — so I always verify the full fix, not just the first thing I find.

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3. Tools Used

- **Active Directory Domain Services (AD DS)** — Windows Server domain controller hosting the simulated environment
- **Active Directory Users and Computers (ADUC)** — inspecting/adjusting OU placement and group membership
- **Group Policy Management Console (GPMC)** — understanding how GPO applies by OU
- **Command line** — `gpupdate /force` to reapply policy during verification
- **Windows client VM** — reproducing and confirming the end-user experience
- **Virtualization** *(VMware / VirtualBox / Hyper-V)* — hosting the lab environment
- **PowerShell** *(planned next step)* — `New-ADUser` script to automate consistent provisioning

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4. Project Timeline

Stage	What I did
1. Build	Stood up a simulated 200-user healthcare AD environment (OUs, users, security groups, GPOs)

2. Intake	Received Ticket NMG-0047 and documented the reported symptoms without jumping to conclusions
3. Investigate	Compared the affected account against a known-good teammate to isolate differences
4. Root cause	Identified two issues via Five Whys analysis: wrong OU and missing security group
5. Resolve	Applied one fix per root cause (OU move, group add, erroneous group removal)
6. Verify	Confirmed both symptoms cleared after `gpupdate /force` and re-login
7. Document	Wrote up the incident and recommended prevention measures

Total resolution time: ~40 minutes.

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5. Key Accomplishments

- Diagnosed and resolved a realistic AD access incident **end-to-end in ~40 minutes**.
- Identified **two distinct root causes** where a less thorough approach would have caught only one — using a known-good baseline comparison instead of guessing.
- Restored the user to full functionality and **verified both symptoms cleared**, not just the obvious one.
- Applied a repeatable troubleshooting method: **investigate → root-cause → fix → verify → document**.
- Recommended prevention: a standardized onboarding checklist plus a **PowerShell (`New-ADUser`) provisioning script** to enforce correct OU and group assignment automatically — turning a one-time fix into a systemic safeguard.

Screenshots

Before

After

Verification